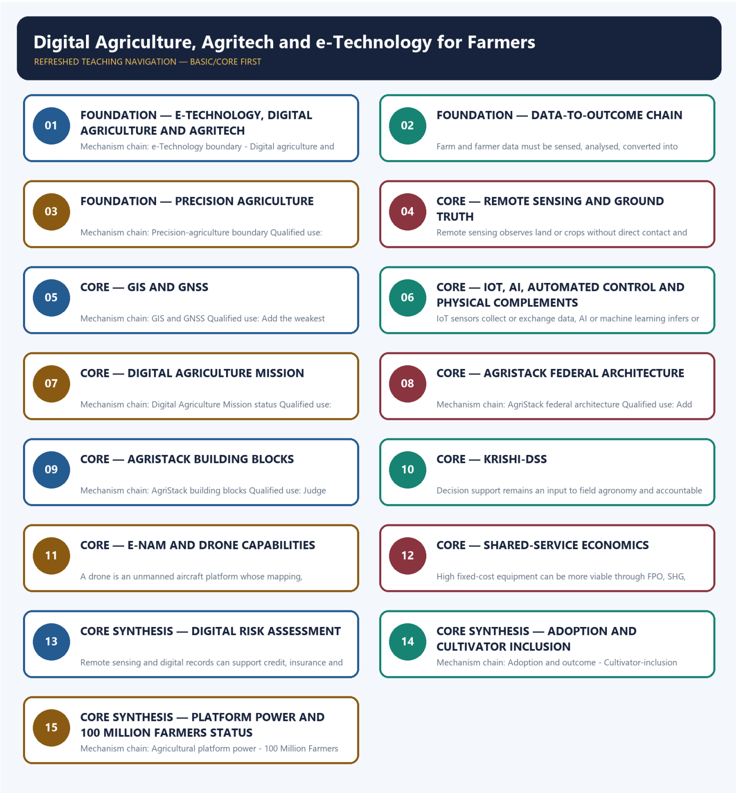
SOLVED PRACTICE WORKBOOK

Digital Agriculture, Agritech and e-Technology for Farmers - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Digital Agriculture, Agritech and e-Technology for Farmers - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies e-Technology boundary?

A. e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. B. Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. C. Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. D. Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare.

Answer: A. Explanation: e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q2. Which option preserves the accounting or regulatory boundary of e-Technology boundary?

A. Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. B. e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. C. Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. D. Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare.

Answer: B. Explanation: e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q3. Which statement uses e-Technology boundary without losing its vintage, basket or legal status?

A. Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. B. GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. C. e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. D. Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions.

Answer: C. Explanation: e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q4. Which option avoids the standard UPSC close-option trap about e-Technology boundary?

A. IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. B. Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. C. GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. D. e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material.

Answer: D. Explanation: e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q5. Which statement correctly identifies Digital agriculture and agritech?

A. Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. B. Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. C. Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. D. Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions.

Answer: A. Explanation: Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q6. Which option preserves the accounting or regulatory boundary of Digital agriculture and agritech?

A. Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. B. Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. C. Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. D. GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct.

Answer: B. Explanation: Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q7. Which statement uses Digital agriculture and agritech without losing its vintage, basket or legal status?

A. Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. B. IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. C. Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. D. GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct.

Answer: C. Explanation: Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q8. Which option avoids the standard UPSC close-option trap about Digital agriculture and agritech?

A. Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. B. GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. C. IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. D. Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal.

Answer: D. Explanation: Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q9. Which statement correctly identifies Data-to-outcome chain?

A. Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. B. GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. C. Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. D. Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction.

Answer: A. Explanation: Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q10. Which option preserves the accounting or regulatory boundary of Data-to-outcome chain?

A. Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. B. Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. C. GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. D. IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others.

Answer: B. Explanation: Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q11. Which statement uses Data-to-outcome chain without losing its vintage, basket or legal status?

A. Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. B. GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. C. Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. D. IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others.

Answer: C. Explanation: Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q12. Which option avoids the standard UPSC close-option trap about Data-to-outcome chain?

A. IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. B. The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. C. Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. D. Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare.

Answer: D. Explanation: Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q13. Which statement correctly identifies Precision-agriculture boundary?

A. Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. B. GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. C. Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. D. IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others.

Answer: A. Explanation: Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q14. Which option preserves the accounting or regulatory boundary of Precision-agriculture boundary?

A. GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. B. Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. C. IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. D. Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement.

Answer: B. Explanation: Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. The other options belong to

different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q15. Which statement uses Precision-agriculture boundary without losing its vintage, basket or legal status?

A. Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. B. The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. C. Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. D. IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others.

Answer: C. Explanation: Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q16. Which option avoids the standard UPSC close-option trap about Precision-agriculture boundary?

A. AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. B. Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. C. The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. D. Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction.

Answer: D. Explanation: Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q17. Which statement correctly identifies Remote sensing and ground truth?

A. Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. B. GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. C. IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. D. Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement.

Answer: A. Explanation: Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q18. Which option preserves the accounting or regulatory boundary of Remote sensing and ground truth?

A. The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. B. Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. C. IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. D. Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement.

Answer: B. Explanation: Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q19. Which statement uses Remote sensing and ground truth without losing its vintage, basket or legal status?

A. Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. B. The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. C. Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. D. AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together.

Answer: C. Explanation: Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q20. Which option avoids the standard UPSC close-option trap about Remote sensing and ground truth?

A. AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. B. The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. C. Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. D. Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions.

Answer: D. Explanation: Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q21. Which statement correctly identifies GIS and GNSS?

A. GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. B. Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. C. IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. D. The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome.

Answer: A. Explanation: GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q22. Which option preserves the accounting or regulatory boundary of GIS and GNSS?

A. AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. B. GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. C. Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. D. The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome.

Answer: B. Explanation: GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q23. Which statement uses GIS and GNSS without losing its vintage, basket or legal status?

A. AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. B. The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. C. GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. D. Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes.

Answer: C. Explanation: GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q24. Which option avoids the standard UPSC close-option trap about GIS and GNSS?

A. AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. B. Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. C. Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. D. GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct.

Answer: D. Explanation: GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q25. Which statement correctly identifies IoT, AI and automated control?

A. IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. B. Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. C. AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. D. The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome.

Answer: A. Explanation: IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q26. Which option preserves the accounting or regulatory boundary of IoT, AI and automated control?

A. The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. B. IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. C. Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. D. AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together.

Answer: B. Explanation: IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q27. Which statement uses IoT, AI and automated control without losing its vintage, basket or legal status?

A. Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. B. Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. C. IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. D. AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together.

Answer: C. Explanation: IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q28. Which option avoids the standard UPSC close-option trap about IoT, AI and automated control?

A. Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. B. Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. C. e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. D. IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others.

Answer: D. Explanation: IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q29. Which statement correctly identifies DPI, platform and physical market?

A. Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. B. Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. C. The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. D. AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together.

Answer: A. Explanation: Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q30. Which option preserves the accounting or regulatory boundary of DPI, platform and physical market?

A. AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. B. Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. C. Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. D. Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes.

Answer: B. Explanation: Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q31. Which statement uses DPI, platform and physical market without losing its vintage, basket or legal status?

A. Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. B. Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. C. Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. D. e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution.

Answer: C. Explanation: Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q32. Which option avoids the standard UPSC close-option trap about DPI, platform and physical market?

A. A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. B. e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. C. Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. D. Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement.

Answer: D. Explanation: Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q33. Which statement correctly identifies Digital Agriculture Mission status?

A. The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. B. Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. C. AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. D. Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority.

Answer: A. Explanation: The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q34. Which option preserves the accounting or regulatory boundary of Digital Agriculture Mission status?

A. e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. B. The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. C. Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. D. Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority.

Answer: B. Explanation: The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q35. Which statement uses Digital Agriculture Mission status without losing its vintage, basket or legal status?

A. A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. B. e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. C. The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. D. Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority.

Answer: C. Explanation: The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q36. Which option avoids the standard UPSC close-option trap about Digital Agriculture Mission status?

A. A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. B. High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. C. e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. D. The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome.

Answer: D. Explanation: The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q37. Which statement correctly identifies AgriStack federal architecture?

A. AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. B. e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. C. Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. D. Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes.

Answer: A. Explanation: AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q38. Which option preserves the accounting or regulatory boundary of AgriStack federal architecture?

A. e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. B. AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. C. Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. D. A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules.

Answer: B. Explanation: AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q39. Which statement uses AgriStack federal architecture without losing its vintage, basket or legal status?

A. A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. B. e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. C. AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. D. High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary.

Answer: C. Explanation: AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q40. Which option avoids the standard UPSC close-option trap about AgriStack federal architecture?

A. A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. B. Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. C. High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. D. AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together.

Answer: D. Explanation: AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q41. Which statement correctly identifies AgriStack building blocks?

A. Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. B. Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. C. A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. D. e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution.

Answer: A. Explanation: Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q42. Which option preserves the accounting or regulatory boundary of AgriStack building blocks?

A. High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. B. Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. C. e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. D. A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules.

Answer: B. Explanation: Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q43. Which statement uses AgriStack building blocks without losing its vintage, basket or legal status?

A. High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. B. A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. C. Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. D. Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route.

Answer: C. Explanation: Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q44. Which option avoids the standard UPSC close-option trap about AgriStack building blocks?

A. Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. B. High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. C. Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. D. Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes.

Answer: D. Explanation: Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q45. Which statement correctly identifies Krishi-DSS boundary?

A. Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. B. A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. C. e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. D. High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary.

Answer: A. Explanation: Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q46. Which option preserves the accounting or regulatory boundary of Krishi-DSS boundary?

A. Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. B. Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. C. A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. D. High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary.

Answer: B. Explanation: Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q47. Which statement uses Krishi-DSS boundary without losing its vintage, basket or legal status?

A. High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. B. Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. C. Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. D. Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence.

Answer: C. Explanation: Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q48. Which option avoids the standard UPSC close-option trap about Krishi-DSS boundary?

A. Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. B. Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. C. Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. D. Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority.

Answer: D. Explanation: Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q49. Which statement correctly identifies e-NAM market completion?

A. e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. B. A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. C. High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. D. Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route.

Answer: A. Explanation: e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q50. Which option preserves the accounting or regulatory boundary of e-NAM market completion?

A. High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. B. e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. C. Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. D. Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route.

Answer: B. Explanation: e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q51. Which statement uses e-NAM market completion without losing its vintage, basket or legal status?

A. Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. B. Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. C. e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. D. Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system.

Answer: C. Explanation: e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q52. Which option avoids the standard UPSC close-option trap about e-NAM market completion?

A. Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. B. Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. C. Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. D. e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution.

Answer: D. Explanation: e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q53. Which statement correctly identifies Drone platform and payload?

A. A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. B. High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. C. Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. D. Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence.

Answer: A. Explanation: A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q54. Which option preserves the accounting or regulatory boundary of Drone platform and payload?

A. Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. B. A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. C. Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. D. Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route.

Answer: B. Explanation: A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q55. Which statement uses Drone platform and payload without losing its vintage, basket or legal status?

A. Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. B. Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. C. A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. D. Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards.

Answer: C. Explanation: A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q56. Which option avoids the standard UPSC close-option trap about Drone platform and payload?

A. Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. B. Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. C. The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved. D. A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules.

Answer: D. Explanation: A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q57. Which statement correctly identifies Shared-service economics?

A. High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. B. Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. C. Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. D. Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence.

Answer: A. Explanation: High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q58. Which option preserves the accounting or regulatory boundary of Shared-service economics?

A. Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. B. High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. C. Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. D. Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system.

Answer: B. Explanation: High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q59. Which statement uses Shared-service economics without losing its vintage, basket or legal status?

A. Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. B. Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. C. High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. D. The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved.

Answer: C. Explanation: High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q60. Which option avoids the standard UPSC close-option trap about Shared-service economics?

A. The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved. B. e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. C. Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. D. High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary.

Answer: D. Explanation: High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q61. Which statement correctly identifies Digital risk assessment?

A. Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. B. Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. C. Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. D. Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards.

Answer: A. Explanation: Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q62. Which option preserves the accounting or regulatory boundary of Digital risk assessment?

A. The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved. B. Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. C. Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. D. Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system.

Answer: B. Explanation: Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q63. Which statement uses Digital risk assessment without losing its vintage, basket or legal status?

A. e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. B. The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved. C. Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. D. Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards.

Answer: C. Explanation: Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q64. Which option avoids the standard UPSC close-option trap about Digital risk assessment?

A. e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. B. The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved. C. Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. D. Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route.

Answer: D. Explanation: Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q65. Which statement correctly identifies Adoption and outcome?

A. Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. B. Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. C. The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved. D. Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system.

Answer: A. Explanation: Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q66. Which option preserves the accounting or regulatory boundary of Adoption and outcome?

A. The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved. B. Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. C. e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. D. Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards.

Answer: B. Explanation: Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q67. Which statement uses Adoption and outcome without losing its vintage, basket or legal status?

A. Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. B. e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. C. Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. D. The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved.

Answer: C. Explanation: Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q68. Which option avoids the standard UPSC close-option trap about Adoption and outcome?

A. Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. B. e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. C. Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. D. Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence.

Answer: D. Explanation: Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q69. Which statement correctly identifies Cultivator-inclusion boundary?

A. Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. B. The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved. C. Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. D. e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material.

Answer: A. Explanation: Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q70. Which option preserves the accounting or regulatory boundary of Cultivator-inclusion boundary?

A. Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. B. Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. C. The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved. D. e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material.

Answer: B. Explanation: Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q71. Which statement uses Cultivator-inclusion boundary without losing its vintage, basket or legal status?

A. Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. B. e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. C. Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. D. Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal.

Answer: C. Explanation: Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q72. Which option avoids the standard UPSC close-option trap about Cultivator-inclusion boundary?

A. Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. B. Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. C. Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. D. Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system.

Answer: D. Explanation: Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q73. Which statement correctly identifies Agricultural platform power?

A. Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. B. e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. C. The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved. D. Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal.

Answer: A. Explanation: Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q74. Which option preserves the accounting or regulatory boundary of Agricultural platform power?

A. Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. B. Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. C. Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. D. e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material.

Answer: B. Explanation: Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q75. Which statement uses Agricultural platform power without losing its vintage, basket or legal status?

A. Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. B. Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. C. Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. D. Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction.

Answer: C. Explanation: Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q76. Which option avoids the standard UPSC close-option trap about Agricultural platform power?

A. Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. B. Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. C. Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. D. Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards.

Answer: D. Explanation: Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q77. Which statement correctly identifies 100 Million Farmers status?

A. The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved. B. e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. C. Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. D. Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal.

Answer: A. Explanation: The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q78. Which option preserves the accounting or regulatory boundary of 100 Million Farmers status?

A. Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. B. The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved. C. Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. D. Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare.

Answer: B. Explanation: The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q79. Which statement uses 100 Million Farmers status without losing its vintage, basket or legal status?

A. Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. B. Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. C. The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved. D. Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions.

Answer: C. Explanation: The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q80. Which option avoids the standard UPSC close-option trap about 100 Million Farmers status?

A. Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. B. GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. C. Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. D. The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved.

Answer: D. Explanation: The World Economic Forum's 100 Million Farmers initiative is a multistakeholder platform with a stated 2030 support ambition, not a Government of India scheme or evidence that the target has been achieved. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

PYQS AND ANSWER PRACTICE

VERIFIED PYQ OWNERSHIP AUDIT

Audited ledgers route the 2023 GS-III e-Technology demand and objective concepts on drone applications and the WEF 100 Million Farmers platform. The Basic session and practice preserve these concepts without inferring unavailable objective keys or platform achievement.

OWNER PYQ LEDGER EXTRACTS

11. PYQ closure

PYQ demand	What a complete answer must cover
2020 Prelims: drone applications including agriculture	Platform/payload distinction; mapping, monitoring and spraying; applications are not universal capabilities
2023 GS-III: how e-technology helps farmers in production and marketing	Stage-wise production uses + market uses + constraints + institutional complements
2025 GS-III: nanotechnology advancements in agriculture and farmer welfare	Input/sensor/delivery applications + productivity/cost/risk channels + safety, access and regulation
Official syllabus: e-technology in aid of farmers	Full data-to-decision-to-action chain, public architecture, inclusion, governance and outcomes

2023 GS-III answer engine - 10 marks / 150 words

Introduction: Define e-technology as digital and electronic systems supporting farm decisions, services and markets.

Production

1. GIS/remote sensing for acreage, crop health, drought and planning.
2. Weather, soil and pest advisories for crop/input decisions.
3. Sensors, precision irrigation/fertigation and drones for targeted action.
4. Digital credit, insurance and loss assessment for risk management.

Marketing

1. price/arrival information and e-NAM discovery;
2. digital assaying, warehouse records and traceability;
3. FPO aggregation, logistics matching and digital settlement.

Critical line: technology requires connectivity, extension, finance, physical infrastructure, data rights and grievance redress.

Conclusion: move from app-centric digitisation to inclusive, interoperable and outcome-accountable digital agriculture.

2025 nanotechnology answer engine - 15 marks / 250 words

1. Define nano-scale applications without claiming automatic superiority.
2. Cover nano-fertilisers/formulations, sensors, diagnostics, targeted delivery and

post-harvest uses.

3. Trace welfare through yield/quality, input efficiency, lower loss and resilience.
4. Add affordability, field evidence, toxicity, regulation, labelling and extension.
5. Conclude with farmer-centric evaluation based on net income and ecological safety.

Demand decoding: The directive **answer** requires a direct position on “11. PYQ closure”, every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the Economy demand in “11. PYQ closure”.

Analytical body:

1. **Claim and named evidence:** 2020 Prelims: drone applications including agriculture Platform/payload distinction; mapping, monitoring and spraying; applications are not universal capabilities **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

2. **Claim and named evidence:** 2023 GS-III: how e-technology helps farmers in production and marketing Stage-wise production uses + market uses + constraints + institutional complements **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
3. **Claim and named evidence:** 2025 GS-III: nanotechnology advancements in agriculture and farmer welfare Input/sensor/delivery applications + productivity/cost/risk channels + safety, access and regulation **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
4. **Claim and named evidence:** Official syllabus: e-technology in aid of farmers Full data-to-decision-to-action chain, public architecture, inclusion, governance and outcomes **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
5. **Claim and named evidence:** Introduction: Define e-technology as digital and electronic systems supporting farm **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
6. **Claim and named evidence:** GIS/remote sensing for acreage, crop health, drought and planning. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

Counter-position / limit: An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

Qualified conclusion: The answer must resolve the Economy demand in “11. PYQ closure”.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

How to improve this answer: For “11. PYQ closure”, replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

Recent PYQ Integration (2024-2025)

> **Status:** 2024-2025 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-PRELIMS-2024-2025.md`. **Answer-key rule:** The official 2024-2025 Prelims Set-A keys are present in the repository and CSAT Set-A keys are supplied; even so, no option or answer is recorded or inferred in this integration.

- **Years represented:** 2024
- **Paper(s):** Prelims GS-I
- **Routed question demands:** 1

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	Prelims GS-I	26	'100 Million Farmers' platform description	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- '100 Million Farmers' platform description

> This block integrates the 2024-2025 examinable demand and paper metadata. It is kept separate from the 2018-2023 block and does not convert an unkeyed/answer-free objective question into a solved answer.

Historical PYQ Integration (2018-2023)

> **Status:** Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-MAINS-GS3-GS4-2018-2023.md`, `_PYQ-ROUTING-PRELIMS-2018-2023.md`. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2020, 2023
- **Paper(s):** GS-III, Prelims GS-I
- **Routed question demands:** 2

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2020	Prelims GS-I	42	Drone applications in agriculture volcano and wildlife research	Objective question; official key unavailable locally	Cross-routed to technical and farmer-economy owners; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2023	GS-III	3	e-Technology helping farmers in agricultural production and marketing	Explain · 10 marks · 150 words	Routed to dedicated e-technology owner	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.

What this owner must now support

- Drone applications in agriculture volcano and wildlife research
- e-Technology helping farmers in agricultural production and marketing

> The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

Historical PYQ Integration (2018-2023)

> **Status:** Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-MAINS-GS3-GS4-2018-2023.md`.

- **Years represented:** 2023
- **Paper(s):** GS-III
- **Routed question demands:** 1

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2023	GS-III	3	e-Technology helping farmers in agricultural production and marketing	Explain · 10 marks · 150 words	Routed to dedicated e-technology owner	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.

What this owner must now support

- e-Technology helping farmers in agricultural production and marketing

> The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

PYQ DEMAND CARD 1 - 2023 GS-III

Demand: How e-Technology helps farmers in agricultural production and marketing.

Status: Official-paper demand routed in the audited 2018-2023 GS-III ledger.

Model solution: Data-to-outcome chain: Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. **Remote sensing and ground truth:** Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. **DPI, platform and physical market:** Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. **e-NAM market completion:** e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. **Shared-service economics:** High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. **Adoption and outcome:** Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. **Cultivator-inclusion boundary:** Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

Demand decoding: The directive **answer** requires a direct position on “PYQ DEMAND CARD 1 - 2023 GS-III”, every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Data-to-outcome chain: Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. **Remote sensing and ground truth:** Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. **DPI, platform and physical market:** Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. **e-NAM market completion:** e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. **Shared-service economics:** High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. **Adoption and outcome:** Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. **Cultivator-inclusion boundary:** Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

Analytical body:

1. **Claim and named evidence:** Demand: How e-Technology helps farmers in agricultural production and marketing. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
2. **Claim and named evidence:** Status: Official-paper demand routed in the audited 2018-2023 GS-III ledger. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

Counter-position / limit: An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

Qualified conclusion: Data-to-outcome chain: Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. **Remote sensing and ground truth:** Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. **DPI, platform and physical market:** Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. **e-NAM market completion:** e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. **Shared-service economics:** High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. **Adoption and outcome:** Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. **Cultivator-inclusion boundary:** Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

How to improve this answer: For “PYQ DEMAND CARD 1 - 2023 GS-III”, replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

ORIGINAL MAINS 1 - 10 MARKS

Question: Explain the data-to-decision-to-action chain in digital agriculture. Answer in about 150 words.

Model thesis: Claim: e-Technology boundary. **Named evidence/example:** e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Digital agriculture and agritech. **Named evidence/example:** Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Data-to-outcome chain. **Named evidence/example:** Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Claim -> named evidence -> analysis -> qualification:

- e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material.
- Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal.
- Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare.

Qualified conclusion: Claim: e-Technology boundary. **Named evidence/example:** e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Digital agriculture and agritech. **Named evidence/example:** Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Data-to-outcome chain. **Named evidence/example:** Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Demand decoding: The directive **explain** requires a direct position on “Explain the data-to-decision-to-action chain in digital agriculture. Answer in about 150…”, every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: e-Technology boundary. **Named evidence/example:** e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Digital agriculture and agritech. **Named evidence/example:** Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Data-to-outcome chain. **Named evidence/example:** Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Analytical body:

1. **Claim and named evidence:** e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
2. **Claim and named evidence:** Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
3. **Claim and named evidence:** Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

Counter-position / limit: An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

Qualified conclusion: Claim: e-Technology boundary. **Named evidence/example:** e-Technology in agriculture uses electronic, digital, communication and information systems to support farm decisions, services and markets; the Core firewall keeps this Basic route exam-complete without relying on optional Advanced material. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Digital agriculture and agritech. **Named evidence/example:** Digital agriculture is data-enabled management across the farm cycle, while agritech also includes firms, devices, mechanisation, biotechnology, nanotechnology and business models; neither is one portal. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Data-to-outcome chain. **Named evidence/example:** Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

How to improve this answer: For "Explain the data-to-decision-to-action chain in digital agriculture. Answer in about 150…", replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Distinguish remote sensing, GIS, GNSS, IoT and AI. Answer in about 150 words.

Model thesis: Claim: Remote sensing and ground truth. **Named evidence/example:** Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** GIS and GNSS. **Named evidence/example:** GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** IoT, AI and automated control. **Named evidence/example:** IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Claim -> named evidence -> analysis -> qualification:

- Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions.
- GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct.

- IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others.

Qualified conclusion: Claim: Remote sensing and ground truth. **Named evidence/example:** Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** GIS and GNSS. **Named evidence/example:** GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** IoT, AI and automated control. **Named evidence/example:** IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Demand decoding: The directive **answer** requires a direct position on “Distinguish remote sensing, GIS, GNSS, IoT and AI. Answer in about 150 words.”, every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Remote sensing and ground truth. **Named evidence/example:** Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** GIS and GNSS. **Named evidence/example:** GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** IoT, AI and automated control. **Named evidence/example:** IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Analytical body:

1. **Claim and named evidence:** Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

2. **Claim and named evidence:** GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
3. **Claim and named evidence:** IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

Counter-position / limit: An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

Qualified conclusion: Claim: Remote sensing and ground truth. **Named evidence/example:** Remote sensing observes land or crops without direct contact and can signal condition, but cloud, resolution, revisit and crop-similarity limits require ground truth and appeal when used for consequential decisions. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** GIS and GNSS. **Named evidence/example:** GIS stores, combines and analyses geographically referenced data, while GNSS or NavIC supplies positioning and navigation signals; mapping analysis and location signals are related but distinct. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** IoT, AI and automated control. **Named evidence/example:** IoT sensors collect or exchange data, AI or machine learning infers or predicts, and automated control acts; a system may combine them but capability in one layer does not prove the others. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

How to improve this answer: For “Distinguish remote sensing, GIS, GNSS, IoT and AI. Answer in about 150 words.”, replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Assess AgriStack and Krishi-DSS as agricultural digital public infrastructure. Answer in about 250 words.

Model thesis: Claim: Digital Agriculture Mission status. **Named evidence/example:** The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** AgriStack federal architecture. **Named evidence/example:** AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** AgriStack building blocks. **Named evidence/example:** Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Krishi-DSS boundary. **Named evidence/example:** Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Cultivator-inclusion boundary. **Named evidence/example:** Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Claim -> named evidence -> analysis -> qualification:

- The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome.
- AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together.
- Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes.
- Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority.
- Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system.

Qualified conclusion: Claim: Digital Agriculture Mission status. **Named evidence/example:** The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** AgriStack federal architecture. **Named evidence/example:** AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** AgriStack building blocks. **Named evidence/example:** Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Krishi-DSS boundary. **Named evidence/example:** Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Cultivator-inclusion boundary. **Named evidence/example:** Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Demand decoding: The directive **assess** requires a direct position on “Assess AgriStack and Krishi-DSS as agricultural digital public infrastructure. Answer in…”, every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Digital Agriculture Mission status. **Named evidence/example:** The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** AgriStack federal architecture. **Named evidence/example:** AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** AgriStack building blocks. **Named evidence/example:** Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Krishi-DSS boundary. **Named evidence/example:** Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Cultivator-inclusion boundary. **Named evidence/example:** Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Analytical body:

1. **Claim and named evidence:** The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
2. **Claim and named evidence:** AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
3. **Claim and named evidence:** Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
4. **Claim and named evidence:** Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
5. **Claim and named evidence:** Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

Counter-position / limit: An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

Qualified conclusion: Claim: Digital Agriculture Mission status. **Named evidence/example:** The Union Cabinet approved the Digital Agriculture Mission in September 2024 as an umbrella architecture; approval and sanctioned outlay are not expenditure, farmer adoption or measured outcome. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** AgriStack federal architecture. **Named evidence/example:** AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** AgriStack building blocks. **Named evidence/example:** Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Krishi-DSS boundary. **Named evidence/example:** Krishi Decision Support System combines geospatial and administrative information for mapping, monitoring and planning. Decision support remains an input to field agronomy and accountable official judgment rather than an autonomous farm-management authority. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Cultivator-inclusion boundary. **Named evidence/example:** Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

How to improve this answer: For "Assess AgriStack and Krishi-DSS as agricultural digital public infrastructure. Answer in…", replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Why does e-NAM require physical market complements? Answer in about 250 words.

Model thesis: Claim: DPI, platform and physical market. **Named evidence/example:** Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** e-NAM market completion. **Named evidence/example:** e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Agricultural platform power. **Named evidence/example:** Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Claim -> named evidence -> analysis -> qualification:

- Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement.
- e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution.
- Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards.

Qualified conclusion: Claim: DPI, platform and physical market. **Named evidence/example:** Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** e-NAM market completion. **Named evidence/example:** e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Agricultural platform power. **Named evidence/example:** Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Demand decoding: The directive **answer** requires a direct position on “Why does e-NAM require physical market complements? Answer in about 250 words.”, every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: DPI, platform and physical market. **Named evidence/example:** Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** e-NAM market completion. **Named evidence/example:** e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Agricultural platform power. **Named evidence/example:** Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Analytical body:

1. **Claim and named evidence:** Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
2. **Claim and named evidence:** e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
3. **Claim and named evidence:** Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

Counter-position / limit: An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

Qualified conclusion: Claim: DPI, platform and physical market. **Named evidence/example:** Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** e-NAM market completion. **Named evidence/example:** e-NAM is operated by SFAC under the agriculture ministry and digitises discovery and transaction functions, but completed trade still needs assaying, aggregation, logistics, settlement and dispute resolution. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Agricultural platform power. **Named evidence/example:** Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis ->

qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

How to improve this answer: For “Why does e-NAM require physical market complements? Answer in about 250 words.”, replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate agritech through adoption, shared-service economics and farmer welfare. Answer in about 300 words.

Model thesis: Claim: Data-to-outcome chain. **Named evidence/example:** Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Precision-agriculture boundary. **Named evidence/example:** Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Drone platform and payload. **Named evidence/example:** A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Shared-service economics. **Named evidence/example:** High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Adoption and outcome. **Named evidence/example:** Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Cultivator-inclusion boundary. **Named evidence/example:** Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Claim -> named evidence -> analysis -> qualification:

- Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare.

- Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction.
- A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules.
- High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary.
- Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence.
- Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system.

Qualified conclusion: Claim: Data-to-outcome chain. **Named evidence/example:** Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Precision-agriculture boundary. **Named evidence/example:** Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Drone platform and payload. **Named evidence/example:** A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Shared-service economics. **Named evidence/example:** High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Adoption and outcome. **Named evidence/example:** Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Cultivator-inclusion boundary. **Named evidence/example:** Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Demand decoding: The directive **evaluate** requires a direct position on “Evaluate agritech through adoption, shared-service economics and farmer welfare. Answer in…”, every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Data-to-outcome chain. **Named evidence/example:** Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Precision-agriculture boundary. **Named evidence/example:** Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Drone platform and payload. **Named evidence/example:** A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Shared-service economics. **Named evidence/example:** High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Adoption and outcome. **Named evidence/example:** Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Cultivator-inclusion boundary. **Named evidence/example:** Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Analytical body:

1. **Claim and named evidence:** Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
2. **Claim and named evidence:** Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
3. **Claim and named evidence:** A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive

channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

4. **Claim and named evidence:** High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
5. **Claim and named evidence:** Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
6. **Claim and named evidence:** Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

Counter-position / limit: An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

Qualified conclusion: Claim: Data-to-outcome chain. **Named evidence/example:** Farm and farmer data must be sensed, analysed, converted into advice, implemented through complementary inputs or services and evaluated through outcomes; data availability alone is not farmer welfare. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Precision-agriculture boundary. **Named evidence/example:** Precision agriculture varies treatment by measured spatial or temporal need; it does not mean zero-input farming or guarantee lower total resource extraction. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Drone platform and payload. **Named evidence/example:** A drone is an unmanned aircraft platform whose mapping, monitoring or spraying capability depends on its payload, configuration, calibration, trained operation and applicable rules. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Shared-service economics. **Named evidence/example:** High fixed-cost equipment can be more viable through FPO, SHG, cooperative or custom-hiring services than universal individual ownership, but recurring demand, scheduling, maintenance and working capital remain necessary. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Adoption and outcome. **Named evidence/example:** Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Cultivator-inclusion boundary. **Named evidence/example:** Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

How to improve this answer: For “Evaluate agritech through adoption, shared-service economics and farmer welfare. Answer in…”, replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design inclusive and accountable digital agriculture for India. Answer in about 300 words.

Model thesis: Claim: DPI, platform and physical market. **Named evidence/example:** Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** AgriStack federal architecture. **Named evidence/example:** AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** AgriStack building blocks. **Named evidence/example:** Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Digital risk assessment. **Named evidence/example:** Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Adoption and outcome. **Named evidence/example:** Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Cultivator-inclusion boundary. **Named evidence/example:** Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Agricultural platform power. **Named evidence/example:** Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Claim -> named evidence -> analysis -> qualification:

- Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement.
- AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together.
- Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes.
- Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route.
- Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence.
- Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system.
- Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards.

Qualified conclusion: Claim: DPI, platform and physical market. **Named evidence/example:** Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** AgriStack federal architecture. **Named evidence/example:** AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** AgriStack building blocks. **Named evidence/example:** Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Digital risk assessment. **Named evidence/example:** Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Adoption and outcome. **Named evidence/example:** Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Cultivator-inclusion boundary. **Named evidence/example:** Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Agricultural platform power. **Named evidence/example:** Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Demand decoding: The directive **answer** requires a direct position on "Design inclusive and accountable digital agriculture for India. Answer in about 300 words.", every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: DPI, platform and physical market. **Named evidence/example:** Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** AgriStack federal architecture. **Named evidence/example:** AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** AgriStack building blocks. **Named evidence/example:** Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Digital risk assessment. **Named evidence/example:** Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Adoption and outcome. **Named evidence/example:** Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Cultivator-inclusion boundary. **Named evidence/example:** Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Agricultural platform power. **Named evidence/example:** Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Analytical body:

1. **Claim and named evidence:** Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
2. **Claim and named evidence:** AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
3. **Claim and named evidence:** Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
4. **Claim and named evidence:** Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
5. **Claim and named evidence:** Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
6. **Claim and named evidence:** Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status,

programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

7. **Claim and named evidence:** Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

Counter-position / limit: An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

Qualified conclusion: Claim: DPI, platform and physical market. **Named evidence/example:** Agricultural DPI supplies interoperable shared rails and a platform coordinates participants, but neither replaces physical assaying, storage, logistics, finance, settlement, extension or enforcement. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** AgriStack federal architecture. **Named evidence/example:** AgriStack is designed as federated Centre-state digital infrastructure rather than one undifferentiated central database, so standards, state operation, data quality and correction must be analysed together. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** AgriStack building blocks. **Named evidence/example:** Farmer Registry, geo-referenced village maps and crop-sown information or Digital Crop Survey are distinct building blocks; an identity record, parcel map and seasonal crop observation are not substitutes. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Digital risk assessment. **Named evidence/example:** Remote sensing and digital records can support credit, insurance and loss assessment, but model error, basis risk, stale records and parcel mismatch require disclosure, field checks and an appeal route. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Adoption and outcome. **Named evidence/example:** Registrations, app downloads, IDs, connected mandis or distributed devices measure activity or availability; adoption, correct use, net income, resilience and resource outcomes require separate evidence. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Cultivator-inclusion boundary. **Named evidence/example:** Land ownership is not a perfect proxy for cultivation, so tenants, sharecroppers and women cultivators may be excluded unless alternative evidence, assisted access and correction are built into the system. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Agricultural platform power. **Named evidence/example:** Digital platforms can reduce search costs and old intermediation while data, ranking, tying and switching costs create new gatekeeping; interoperability and portability are economic safeguards. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

How to improve this answer: For “Design inclusive and accountable digital agriculture for India. Answer in about 300 words.”, replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.